

Cheaper, Easier CFL Downlighting

The use of downlights in the residential market is increasing, yet energy-efficient CFL downlights have not been widely accepted or adopted by residential builders and homeowners. That's because residential CFL downlights are costly, difficult to install, and not widely available. In addition, existing residential products are not as efficient or reliable as their commercial-market cousins. As a result, builders often install inefficient incandescent lights, increasing electricity bills for consumers and adding to peak demand on electric utilities.

A low-cost, high-performance downlighting system has been developed for residential kitchens. Kitchen lighting is the largest user of lighting energy in California homes; it accounts for one-fourth of the operating cost of residential lighting. The Kitchen Lighting System (KLS), developed at the California Lighting Technology Center (CLTC) with funding from the California Energy Commission's Public Interest Energy Research (PIER) program, features two CFLs that can be controlled from a single ballast, uses fixtures that emit more than 50 lumens per watt, and costs less than half as much to install as current commercial-grade products. The KLS is now offered as a product by Lithonia Lighting under the name CKP62 (see photo). Lithonia has also applied the concepts to two new CFL products for commercial applications—the CCR62 and the CCR82—which also feature two fixtures that can be controlled from a single ballast. Other lighting manufacturers—including Cooper Lighting, Progress Lighting, and Seagull Lighting—have recently introduced similar products for the residential market that incorporate some of the KLS features—proof that a real market transformation is taking place.

Features and Benefits

The KLS incorporates a number of features that help it deliver high performance at low cost, including a thermally enhanced ballast configuration. The KLS



Builders can install the Kitchen Lighting System, sold by Lithonia under the name CKP62, the same way that they would a conventional downlight.

ballast is connected to the main metallic housing for the downlight pan itself, helping to keep the ballast at acceptable operating temperatures and ensuring long lamp life—even when the fixture is buried under insulation or installed in a hot attic space.

The KLS also features a one-ballast, two-fixture design. This approach reduces material and installation costs. The biggest material cost for CFL downlighting is the electronic ballast. The KLS replaces the two one-lamp ballasts used in a conventional approach with one two-lamp ballast. Because two-lamp ballasts cost no more than one-lamp ballasts, this is a significant advantage. Using a single ballast for multiple lamps also cuts the number of hard-wired connections in half, resulting in savings on installation labor.

The KLS is also institutionally transparent and builder-friendly. Builders can install the KLS in much the same way as they would install a conventional downlight. After the first test installations at a series of homes in the Sacramento Municipal Utility District (SMUD), installers reported that the KLS was easier to put in than conventional systems. Its plug-and-play wire connections, includ-

ing a flexible and removable connector design, simplify the wiring to the fixture and cut installation costs. Accessible ballasts, plug-and-play design, and an 11-ft-long wiring harness that allows easy connection with other fixtures combine to make maintenance easier as well.

The reflector optics for the KLS are based on existing commercial-grade CFL products that maximize output while minimizing glare. The CFLs included with the KLS are 26W high-quality, high-output lamps. The ballast chosen for the KLS is a rapid-start unit approved for residential applications (FCC Class B). These features combine to make the KLS far more cost-effective than incandescent and CFL downlight alternatives (see Table 1). The system cuts energy use by 75% with negligible cost increase compared to standard incandescent downlighting, all while providing higher, more uniform illumination levels.

Beyond the Kitchen

The KLS was developed specifically for new residential kitchens, but the system can be installed in any residential space where downlights are used. Its fea-

tures are based on detailed analysis of current construction techniques and feedback from home builders. In addition, the system is appropriate for many different residential applications and can be used as a retrofit. The PIER program is funding a project to optimize the design for the retrofit market.

In October 2005, California's Title 24 required that at least half of the installed wattage in a kitchen be hard-wired high-efficiency sources and fixtures. This requirement effectively eliminates the use of incandescent downlights and screw-based CFLs in kitchens. The KLS will help builders to comply with this stringent standard.

Table 1. Performance of KLS

	Standard Incandescent Lamp	Standard CFL	KLS
Number of downlights	10	10	6
Power per downlight (watts)	65	15	26
Total light output (Lm)	6,080	3,900	7,800
Total power (watts)	650	150	180
Total installed cost (\$)	500	650	420
Operating cost (\$/yr)	76	18	21
Cost increment (\$)	N/A	150	-80
Savings (\$/yr)	N/A	58	55
Simple payback (yr)	N/A	2.6	Immediate

Future Plans

A series of field tests have shown that the KLS performs as expected. SMUD has integrated the KLS into its Advantage Homes program, which encourages home builders to adopt energy-saving devices in new homes. For the next phase of development, researchers at CLTC have started working to enhance the thermal management capabilities in real-world applications. They are also working with Lithonia and other downlight manufacturers to broaden the applicability of the concept by developing a single-fixture version of the product that incorporates the thermal management capabilities of the original version.



This article was adapted from a California Energy Commission PIER Technical Brief produced by E Source.

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To learn more about this and other lighting products, go to www.energy.ca.gov/pier.